

Geometric Field Equations and Riemannian Structure of State-Dependent Interaction Dynamics

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Abstract

We extend the geometric formulation of two-agent interaction dynamics by transitioning from a purely time-dependent model to a state-dependent framework. By introducing an interaction rule where the transformation depends on both time and the active interaction state, $\gamma = \gamma(t, I_S)$, we induce a spatially varying Riemannian metric

$$G(t, I_S) = I + J^T J$$

on the interaction manifold. We derive the Levi-Civita connection and demonstrate that the resulting Ricci curvature arises from two distinct mechanisms: temporal deformation of the interaction metric and spatial variation of the interaction rule.

Building upon this geometric foundation, we introduce an interaction action functional and prove, under the axioms of locality, covariance, and second-order field equations, that the admissible action is uniquely determined by the scalar curvature. This variational approach yields a non-trivial interaction field equation of the form

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu},$$

which is structurally analogous to the Einstein field equations. This framework provides a rigorous nonlinear description of interaction dynamics, where the interaction source tensor $T_{\mu\nu}$ encodes reinforcement and perceptual structures, offering a unified perspective linking differential geometry with behavioral feedback systems.

1 Introduction

Mathematical descriptions of interacting systems often assume that the coupling between variables depends only on time. However, many real interaction systems exhibit state-dependent coupling, where the interaction rule changes depending on the current interaction state. Examples include:

- nonlinear feedback in dynamical systems

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- adaptive behavior in social interactions
- saturation effects in biological responses

In such systems the transformation between interaction variables is not purely time dependent but varies with the state itself. To incorporate this feature, we extend the interaction rule from

$$\gamma = \gamma(t)$$

to

$$\gamma = \gamma(t, I_S)$$

where I_S represents the active interaction variable. This extension introduces spatial variation in the interaction structure and naturally leads to a state-dependent Riemannian metric on the interaction manifold.[2][1]

2 Interaction Variables

We consider two interaction vectors

$$I_S(t) \in \mathbb{R}^2$$

$$I_P(t) \in \mathbb{R}^2$$

representing active and passive interaction components.
We impose the interaction constraint

$$I_P = \gamma(t, I_S)I_S$$

where

$$\gamma(t, I_S) \in \mathbb{R}^{2 \times 2}$$

is a differentiable matrix-valued function depending on both time and interaction state.

2.1 Extended Euclidean Construction

We begin with the extended coordinate space

$$(t, I_P^1, I_P^2, I_S^1, I_S^2)$$

equipped with the Euclidean line element

$$ds^2 = dt^2 + dI_P^T dI_P + dI_S^T dI_S$$

The interaction manifold is obtained by imposing the constraint

$$I_P = \gamma(t, I_S)I_S$$

2.2 Differential of the Interaction Constraint

Taking the differential gives

$$dI_P = \gamma dI_S + (\partial_t \gamma) I_S dt + (\partial_{I_S} \gamma) (dI_S) I_S$$

This shows that the transformation now varies with both time and interaction state. To define the spatial metric, we consider variations at fixed time $dt = 0$. Thus

$$dI_P = \gamma dI_S + (\partial_{I_S} \gamma) (dI_S) I_S$$

Define the Jacobian matrix of the interaction mapping

$$J(t, I_S) = \gamma(t, I_S) + (\nabla_{I_S} \gamma(t, I_S)) I_S$$

where

$$(\nabla_{I_S} \gamma) I_S = \frac{\partial \gamma}{\partial I_S^k} I_S^k$$

2.3 Interaction Metric

Substituting into the Euclidean line element gives

$$ds^2 = dt^2 + dI_S^T G(t, I_S) dI_S$$

where the interaction metric is

$$G(t, I_S) = \mathbf{I} + J(t, I_S)^T J(t, I_S)$$

where

$$J = \gamma + (\nabla_{I_S} \gamma) I_S$$

Properties:

- symmetric
- positive definite
- state dependent

Thus the interaction manifold is

$$M = \mathbb{R}_t \times \mathbb{R}_{I_S}^2$$

with metric

$$g_{\mu\nu} = \begin{bmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{bmatrix},$$

3 Interaction Flow Tensor

Define the interaction flow tensor

$$K_j^i = \frac{1}{2} G^{ik} \partial_t G_{kj}$$

which measures the temporal deformation of the interaction metric. This tensor represents the rate at which the interaction geometry evolves in time. This tensor plays a role analogous to the extrinsic curvature in geometric evolution systems.

4 Levi-Civita Connection

Because G_{ij} depends on both t and I_S , the non-vanishing Christoffel symbols are Temporal terms

$$\begin{aligned} \Gamma_{ij}^0 &= -\frac{1}{2a^2} \partial_t G_{ij} \\ \Gamma_{0j}^i &= K_j^i \end{aligned}$$

Spatial terms

$$\Gamma_{jk}^i = \frac{1}{2} G^{il} (\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk})$$

The spatial terms arise purely from the state dependence of the interaction rule.

Theorem 4.1 (Connection structure). For the metric

$$g_{\mu\nu} = \begin{bmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{bmatrix}$$

the non-vanishing Christoffel symbols are

$$\begin{aligned} \Gamma_{ij}^0 &= -\frac{1}{2a^2} \partial_t G_{ij} \\ \Gamma_{0j}^i &= K_j^i \\ \Gamma_{jk}^i &= \frac{1}{2} G^{il} (\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk}) \end{aligned}$$

Proof. The Christoffel symbols are defined by [3]

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$$

For the metric

$$g_{\mu\nu} = \begin{bmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{bmatrix}$$

we have

$$g_{00} = a^2, g_{0i} = 0, g_{ij} = G_{ij}(t, I_S)$$

Since g_{00} is constant,

$$\partial_\mu g_{00} = 0$$

Thus

$$\Gamma_{ij}^0 = \frac{1}{2}g^{00}(-\partial_0 g_{ij}) = -\frac{1}{2a^2}\partial_t G_{ij}$$

For mixed components,

$$\Gamma_{0j}^i = \frac{1}{2}G^{ik}\partial_t G_{kj}$$

which defines

$$K_j^i = \frac{1}{2}G^{ik}\partial_t G_{kj}$$

Finally, because G_{ij} depends on I_S

$$\Gamma_{jk}^i = \frac{1}{2}G^{il}(\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk})$$

All other components vanish. □

5 Ricci Curvature Structure

The Ricci tensor now decomposes into two parts:

$$R_{ij} = R_{ij}^{\text{time}} + R_{ij}^{\text{space}}$$

where Temporal curvature:

$$R_{ij}^{\text{time}} = \dot{K}_{ij} + K_{ik}K_j^k + K_{ij}\text{Tr}(K)$$

Spatial curvature:

$$R_{ij}^{\text{space}}$$

is generated by the spatial Levi-Civita connection from $\partial_k G_{ij}$

Theorem 5.1 (Ricci Tensor Decomposition). For the interaction manifold (M, g) with metric

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{pmatrix},$$

where $G_{ij}(t, I_S) = \delta_{ij} + [J(t, I_S)]^T J(t, I_S)$ and $J(t, I_S) = \gamma(t, I_S) + (\nabla_{I_S} \gamma)I_S$, the Ricci tensor $R_{\mu\nu}$ decomposes as

$$R_{ij} = \dot{K}_{ij} + K_{ik}K_j^k + K_{ij}\text{Tr}(K) + R_{ij}^{\text{space}},$$

where $K^i_j = \frac{1}{2}G^{ik}\partial_t G_{kj}$ is the interaction flow tensor, the dot denotes the time derivative, and R_{ij}^{space} is the Ricci tensor of the spatial metric G_{ij} computed with respect to its own Levi-Civita connection.

Theorem 5.2 (Temporal Component). The (0,0)-component of the Ricci tensor is given by

$$R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2).$$

Part 1: Computation of R_{00} (Theorem 5.2).

Proof. The Ricci tensor is defined by [3]

$$R_{\mu\nu} = \partial_\lambda \Gamma_{\mu\nu}^\lambda - \partial_\nu \Gamma_{\mu\lambda}^\lambda + \Gamma_{\mu\sigma}^\lambda \Gamma_{\lambda\nu}^\sigma - \Gamma_{\mu\lambda}^\sigma \Gamma_{\sigma\nu}^\lambda.$$

From Theorem 4.1, the only non-vanishing Christoffel symbols are:

Temporal:

$$\Gamma_{ij}^0 = -\frac{1}{2a^2}\partial_t G_{ij}, \quad \Gamma_{0j}^i = K^i_j,$$

Spatial:

$$\Gamma_{jk}^i = \frac{1}{2}G^{il}(\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk}).$$

Substitute into the Ricci formula for $\mu = \nu = 0$:

$$R_{00} = \partial_\lambda \Gamma_{00}^\lambda - \partial_0 \Gamma_{0\lambda}^\lambda + \Gamma_{0\sigma}^\lambda \Gamma_{\lambda 0}^\sigma - \Gamma_{0\lambda}^\sigma \Gamma_{\sigma 0}^\lambda.$$

Since $g_{00} = a^2$ is constant, $\Gamma_{00}^\lambda = 0$ for all λ . Thus $\partial_\lambda \Gamma_{00}^\lambda = 0$.

The second term is $-\partial_0 \Gamma_{0\lambda}^\lambda = -\partial_t(\Gamma_{0i}^i) = -\partial_t \text{Tr}(K)$, because $\Gamma_{0\lambda}^0 = 0$. The quadratic terms simplify using the symmetry of K_{ij} :

$$\Gamma_{0\sigma}^\lambda \Gamma_{\lambda 0}^\sigma - \Gamma_{0\lambda}^\sigma \Gamma_{\sigma 0}^\lambda = K^i_j K^j_i = \text{Tr}(K^2).$$

(All contributions involving Γ^0 vanish due to the block-diagonal structure of the metric.)

Combining these yields

$$R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2).$$

This completes the proof of Theorem 5.2. □

Part 2: Computation of R_{ij} (Theorem 5.1).

Proof. For the spatial components, we compute each term in the Ricci formula carefully. The full expression for R_{ij} receives contributions from:

Temporal deformation terms (involving ∂_t of the metric): The time derivative of the relevant Christoffel symbols produces $\dot{K}_{ij}$. The quadratic Christoffel contractions involving one temporal and one spatial index yield the terms $K_{ik}K^k_j$ and $K_{ij} \text{Tr}(K)$.

Purely spatial terms: These are exactly the Ricci tensor R_{ij}^{space} computed from the spatial Christoffel symbols Γ_{jk}^i with respect to the spatial metric G_{ij} :

$$R_{ij}^{\text{space}} = \partial_k \Gamma_{ij}^k - \partial_j \Gamma_{ik}^k + \Gamma_{il}^k \Gamma_{jk}^l - \Gamma_{jl}^k \Gamma_{ik}^l.$$

When all terms are collected, the cross terms involving both temporal and spatial connections produce the interaction acceleration tensor

$$E_{ij} := \dot{K}_{ij} + K_{ik}K_j^k,$$

while the trace term $K_{ij} \text{Tr}(K)$ arises from the contraction with the temporal flow. Adding the intrinsic spatial curvature gives the complete decomposition

$$R_{ij} = \dot{K}_{ij} + K_{ik}K_j^k + K_{ij} \text{Tr}(K) + R_{ij}^{\text{space}}.$$

Remark. The decomposition cleanly separates curvature induced by temporal evolution of the interaction rule ($\dot{K}$ terms) from curvature induced by state-dependent spatial variation (R_{ij}^{space}). This geometric splitting is central to the interpretation of emotional escalation and nonlinear feedback in the interaction manifold. This completes the proof of Theorem 5.1. $\square$

6 Emotional Quantity Tensor

From a Ricci tensor

$$R_{ij} = E_{ij} + K_{ij} \text{Tr}(K) + R_{ij}^{\text{space}}$$

We define the intrinsic interaction acceleration

$$E_{ij} = \dot{K}_{ij} + K_{ik}K_j^k$$

The tensor E_{ij} corresponds to the intrinsic interaction acceleration component of the Ricci tensor. This quantity describes nonlinear interaction acceleration independent of collective coupling.

Curvature in the interaction manifold arises from two mechanisms:

1. Temporal deformation : changes of interaction rules over time.
2. State-dependent coupling : changes of interaction rules depending on the interaction intensity itself.

The second mechanism introduces nonlinear feedback in the geometry.[2][4]

7 State Term I_S

Introducing I_S into γ is essential for three reasons.[]

7.1 Real interaction systems are state dependent

In many systems the interaction strength depends on the current interaction level. Examples:[4]

- emotional escalation
- biological response saturation
- adaptive feedback systems

Thus

$$\gamma = \gamma(t, I_S)$$

captures nonlinear interaction feedback.

7.2 Mathematical completeness of the geometry

If $\gamma = \gamma(t)$ then

$$\partial_k G_{ij} = 0$$

and the spatial geometry becomes flat. Allowing I_S dependence generates non-trivial spatial curvature.

7.3 Toward interaction field equations

State-dependent metrics allow curvature structures similar to those in geometric field theories. This opens the possibility of formulating interaction dynamics in the form

$$R_{\mu\nu} = \text{interaction source tensor}$$

analogous to geometric field equations.

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = T_{\mu\nu}^{\text{Reward}}$$

This equation provides a geometric interaction field equation analogous to Einstein-type field equations.

8 Interaction Action Functional and Field Equation

To obtain a geometric field equation governing interaction dynamics, we introduce an action functional defined on the interaction manifold. Let the interaction manifold be

$$M = \mathbb{R}_t \times \mathbb{R}_{I_S}^2$$

equipped with the metric

$$g_{\mu\nu} = \begin{bmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{bmatrix}$$

where

$$G_{ij}(t, I_S) = \mathbf{I} + J(t, I_S)^T J(t, I_S)$$

and

$$J = \gamma + (\nabla_{I_S} \gamma) I_S.$$

8.1 Axiom of the Interaction Action Derivation (Physical-Geometric Requirements)

Axiom 8.1 (Locality). The interaction dynamics at a point depend only on finite derivatives of the metric at that point. Hence the action has form

$$S[g] = \int_M L(g, \partial g, \partial^2 g, \dots) d\mu_g.$$

where L is a scalar density constructed locally from $g_{\mu\nu}$.

Axiom 8.2 (Covariance). Interaction laws are invariant under smooth coordinate transformations

$$x^\mu \mapsto x'^\mu(x).$$

Therefore:

$$S[g] = S[\phi^* g]$$

for every diffeomorphism $\phi : M \rightarrow M$.

Axiom 8.3 (Metric Dynamics). The metric $g_{\mu\nu}$ is the only dynamical geometric field. Thus the Lagrangian depends solely on geometric tensors generated from g .

Axiom 8.4 (Second-Order Field Equations). Euler-Lagrange equations must contain at most second derivatives of the metric. This requirement guarantees well-posed interaction evolution and excludes higher-order instabilities.

8.2 Consequence of Covariance

Lemma 8.1 (Scalar Construction). Any diffeomorphism-invariant local Lagrangian must be constructed from contractions of tensors built from:

$$g_{\mu\nu}, R_{\sigma\mu\nu}^p, \nabla_\alpha R_{\sigma\mu\nu}^p, \dots$$

Proof. Under coordinate change: partial derivatives are not tensors, curvature tensors are. Thus only tensorial contractions produce invariant scalars. $\square$

Hence

$$L = f(g, R, \nabla R, \nabla^2 R, \dots).$$

8.3 Restriction from Second-Order Dynamics

Lemma 8.2. If the Lagrangian contains derivatives of curvature, then field equations contain derivatives higher than second order.

Proof. Curvature already contains second derivatives:

$$R \sim \partial^2 g.$$

If L contains

$$\nabla R$$

,

$$\nabla R \sim \partial^3 g.$$

Variation produces

$$\delta S \sim \partial^4 g.$$

Hence equations become fourth order. This violates Axiom 4. □

Corollary 8.1. The admissible Lagrangian depends only on curvature algebraically:

$$L = L(g, R_{\mu\nu\rho\sigma}).$$

8.4 Classification of Scalar Invariants

We now classify all scalar invariants constructed algebraically from curvature. In dimension $n = 3$ (interaction manifold dimension), the independent scalar invariants are generated by contractions of:

$$R_{\mu\nu\rho\sigma}.$$

But in 3D geometry:

Theorem 8.1 (3D Curvature Identity). The Riemann tensor satisfies

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} - g_{\mu\sigma}R_{\nu\rho} - g_{\nu\rho}R_{\mu\sigma} + g_{\nu\sigma}R_{\mu\rho} - \frac{R}{2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}).$$

Hence it is completely determined by Ricci tensor.

Therefore every scalar invariant reduces to a function of

$$R, R_{\mu\nu}R^{\mu\nu}.$$

8.5 Second-Order Constraint Eliminates Higher Powers

Lemma 8.3. Quadratic curvature invariants generate fourth-order equations.

Proof. Consider

$$S = \int R_{\mu\nu} R^{\mu\nu} \sqrt{|g|}.$$

Variation produces terms:

$$\delta R_{\mu\nu} \sim \nabla \nabla \delta g,$$

giving fourth derivatives of g . Thus equations exceed second order. $\square$

Hence quadratic invariants are forbidden.

8.6 Interaction Field Equation

Theorem 8.2 (Uniqueness of the Interaction Action Functional). Let (M, g) be a 3-dimensional interaction manifold with coordinates (t, I_S^1, I_S^2) equipped with the metric

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{pmatrix},$$

where $G_{ij}(t, I_S) = \delta_{ij} + J^T J$ is the induced spatial metric. Assume the following axioms:

- Axiom 1 (Locality): The dynamics at each point depend only on the metric and its derivatives up to finite order at that point.
- Axiom 2 (Diffeomorphism Covariance): The action is invariant under smooth coordinate transformations.
- Axiom 3 (Metric Dynamics): The metric $g_{\mu\nu}$ is the sole dynamical field.
- Axiom 4 (Second-Order Field Equations): The Euler-Lagrange equations derived from the action contain at most second derivatives of the metric.

Then, up to a boundary term, the unique admissible action functional is

$$S[g] = \int_M \left(\frac{1}{2\kappa} R + \Lambda \right) \sqrt{|g|} d^3x,$$

where R is the Ricci scalar, κ is the coupling constant, and Λ is the cosmological constant. Variation of this action yields the interaction field equation

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Proof. 1. **Locality and Covariance.** By locality, the Lagrangian density $\mathcal{L}$ is a scalar function constructed from the metric $g_{\mu\nu}$ and its derivatives. Diffeomorphism covariance requires $\mathcal{L}$ to be built from tensorial quantities. Since the Christoffel symbols $\Gamma_{\mu\nu}^\lambda$ are not tensors, the lowest-order tensorial objects available are the Riemann curvature tensor $R^\rho_{\sigma\mu\nu}$ and its covariant derivatives. Thus, any admissible $\mathcal{L}$ must take the form

$$\mathcal{L} = f(g_{\mu\nu}, R^\rho_{\sigma\mu\nu}, \nabla_\alpha R^\rho_{\sigma\mu\nu}, \nabla_\alpha \nabla_\beta R^\rho_{\sigma\mu\nu}, \dots).$$

2. **Dimensional Reduction in 3 Dimensions.** In three spacetime dimensions, the Weyl tensor vanishes identically, and the Riemann tensor is completely determined by the Ricci tensor $R_{\mu\nu}$ and the scalar curvature R via the identity

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} - g_{\mu\sigma}R_{\nu\rho} - g_{\nu\rho}R_{\mu\sigma} + g_{\nu\sigma}R_{\mu\rho} - \frac{R}{2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}).$$

Consequently, every scalar invariant constructed algebraically from the Riemann tensor reduces to a function of the independent scalars R and $R_{\mu\nu}R^{\mu\nu}$. Higher-order invariants involving covariant derivatives of the curvature are possible in principle but will be constrained by the next axiom.

3. **Second-Order Constraint.** The curvature tensor $R_{\mu\nu\rho\sigma}$ itself contains second derivatives of the metric. If the Lagrangian $\mathcal{L}$ contains terms involving ∇R or higher, the variation δS will produce terms with third and fourth derivatives of $g_{\mu\nu}$, leading to fourth-order (or higher) field equations. This violates Axiom 4. Moreover, quadratic curvature invariants such as $\int R_{\mu\nu}R^{\mu\nu} \sqrt{|g|} d^3x$ also generate fourth-order equations. To see this explicitly, note that the variation of the Ricci tensor satisfies

$$\delta R_{\mu\nu} \sim \nabla_\lambda \nabla_\sigma (\delta g^{\lambda\sigma}) + \dots,$$

so the Euler-Lagrange equations derived from quadratic terms contain fourth derivatives of the metric. Such terms are therefore excluded.

4. **Admissible Lagrangians.** The only scalar densities satisfying all four axioms are linear combinations of the volume element and the scalar curvature:

$$\mathcal{L} = \alpha + \beta R,$$

where α, β are constants. The unique diffeomorphism-invariant volume element on (M, g) is $\sqrt{|g|} d^3x$; no other scalar density satisfies covariance.

5. **Normalization and Variation.** Setting $\beta = 1/(2\kappa)$ and $\alpha = \Lambda$ (after absorbing the conventional factor), the action becomes

$$S[g] = \int_M \left(\frac{1}{2\kappa} R + \Lambda \right) \sqrt{|g|} d^3x.$$

The Euler-Lagrange variation of the Einstein-Hilbert term $\int R\sqrt{|g|}$ is well-known to yield the Einstein tensor:

$$\frac{2}{\sqrt{|g|}} \frac{\delta}{\delta g^{\mu\nu}} \left(\int R\sqrt{|g|} \right) = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}.$$

Adding the cosmological term produces the desired field equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu},$$

where the interaction source tensor $T_{\mu\nu}$ arises from possible matter or interaction contributions (to be specified in applications).

6. Uniqueness. Any other diffeomorphism-invariant, local scalar density leading to second-order equations must reduce to the above form in three dimensions. Hence the action is unique up to boundary terms and the choice of constants κ and Λ .

□

Variation yields an interaction field equation, i.e.,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

which coincides with the interaction field equation previously introduced — but now derived, not assumed.

8.7 Non-Triviality of the Interaction Geometry

A potential issue in geometric interaction models is the emergence of trivial solutions where the curvature vanishes identically. We show that under the state-dependent metric construction, the geometry is generically non-trivial.

Theorem 8.3 (Non-trivial curvature condition). Let the spatial metric be

$$G_{ij}(t, I_S) = \mathbf{I} + J^T J$$

with

$$J = \gamma(t, I_S) + (\nabla_{I_S} \gamma(t, I_S)) I_S.$$

If the interaction rule satisfies

$$\nabla_{I_S} \gamma \neq 0$$

in some region of the interaction space, then the spatial curvature R_{ij}^{space} is generically non-zero.

Proof. The spatial curvature tensor depends on spatial derivatives of the metric

$$\partial_k G_{ij}.$$

Since

$$G = \mathbf{I} + J^T J$$

we obtain

$$\partial_k G = (\partial_k J)^T J + J^T (\partial_k J).$$

If $\nabla_{I_S} \gamma \neq 0$ then $\partial_k \neq 0$. Therefore

$$\partial_k G_{ij} \neq 0$$

which implies that the spatial Christoffel symbols Γ_{jk}^i are non-zero. Non-vanishing Christoffel symbols generate non-zero spatial Ricci curvature R_{ij}^{space} .

Hence the interaction geometry is generically curved and cannot reduce to a trivial flat structure except in the special case

$$\nabla_{I_S} \gamma = 0.$$

□

8.8 Special Case: Trivial Geometry

Trivial curvature occurs only if

$$\gamma = \gamma(t)$$

so that

$$\nabla_{I_S} \gamma = 0.$$

In this case

$$G_{ij}(t) = \mathbf{I} + \gamma^T \gamma$$

becomes spatially homogeneous and

$$R_{ij}^{\text{space}} = 0.$$

Thus the curvature reduces purely to temporal deformation. This corresponds exactly to the earlier interaction model without state-dependent coupling.

9 Conclusion

This paper has developed a comprehensive geometric formulation for continuous interaction dynamics by extending the interaction trajectory $\gamma(t)$ into a generalized interaction manifold parameterized by both time t and interaction-state coordinates I_S . The central contribution of this work is the introduction of a state-dependent interaction metric $G_{ij}(t, I_S)$, which formalizes the geometry of the interaction space. Unlike earlier models that rely solely on temporal evolution, this framework explicitly incorporates the internal interaction state, allowing the system to capture nonlinear variations in perceptual sensitivity and interaction intensity.

A key theoretical advancement presented in this study is the formal derivation of the interaction field equation through a variational principle. By defining an interaction action functional $S[g]$ and applying the axioms of locality, covariance, and second-order dynamics, we have shown that the geometry is governed by a unique set of field equations. The resulting structure,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

, provides a robust mathematical bridge between differential geometry and behavioral science, where the interaction source tensor $T_{\mu\nu}$ accounts for reinforcement effects and perceptual forcing.

Furthermore, we have demonstrated that the introduction of state-dependence ($\nabla_{I_S}\gamma \neq 0$) ensures the non-triviality of the interaction geometry, preventing the manifold from collapsing into a flat Euclidean space. This ensures that the model captures meaningful dynamical structures inherent in complex adaptive systems. The proposed framework thus provides a unified geometric perspective linking interaction dynamics, perceptual sensitivity structures, and reinforcement mechanisms within a single, rigorous formalism. Future research will focus on extending this manifold to higher-dimensional spaces and exploring empirical validations within cognitive and social interaction systems.

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